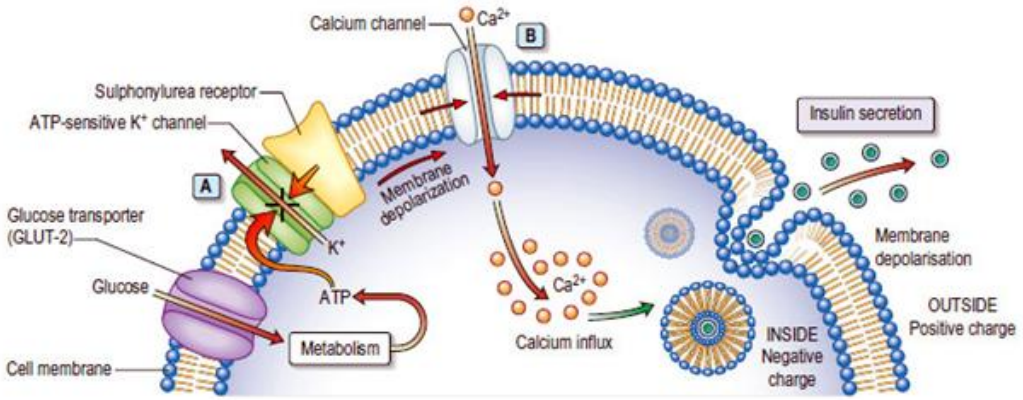
ยารักษาโรคเบาหวาน

(ANTIDIABETIC DRUGS)

Asst.Prof.Dr.Somchai Sriwiriyajan

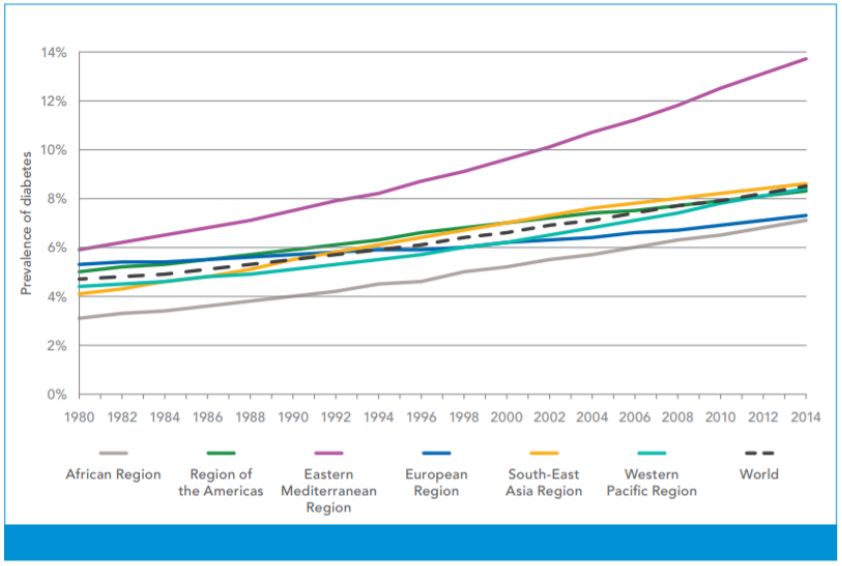
Pharmacology, PSU



โรคเบาหวาน
(DIABETES MELLITUS, DM)

- **Diabetes = a passing through, mellitus = honey**
- **Metabolic disorder characterized by a significantly raised blood glucose**
- **A major public health problem**
- **WHO :**
  - **1985 : 30 million**
  - **2025 : 589 million**
  - **2045 : >729 million**

FIGURE 4B. TRENDS IN PREVALENCE OF DIABETES, 1980–2014, BY WHO REGION



- **In Thailand ~ 6.36 million** (International Diabetes Federation, 2024)

สาเหตุและพยาธิสภาพของโรคเบาหวาน

ขาดอินสุลิน, ขาดอินสุลินรีเซพเตอร์, มีความผิดปกติในระดับเซลล์

กลูโคสเข้าเซลล์ไม่ได้

Chronic Hyperglycemia

AGE formation
Glucose autoxidation
Glucosamine
Oxidative phosphorylation
→ ROS

Oxidative stress

Low capacity of antioxidant enzymes
- Superoxide dismutase
- Catalase
- Glutathione peroxidase

β-cell dysfunction

เกิดภาวะแทรกซ้อน เช่น ติดเชื้อง่าย
ความดันโลหิตสูง หลอดเลือดอุดตัน ตาพร่ามัว
ประสาทพิการ ไตวาย

Glycosuria (Polyuria)

Dehydration (Polydypsia)

Polyphagia

ร่างกายขาดพลังงาน

การเผาผลาญโปรตีนและไขมันเพิ่มขึ้น

กล้ามเนื้ออ่อนแรง

ketone body
- acetoacetic acid
- β-hydroxybutyric acid

ketoacidosis

Diagnosing diabetes

The American Diabetes Association (ADA) criteria

- **Fasting blood glucose level ≥ 126 mg/dL (7.0 mmol/L)**
- **2-hour oral glucose tolerance test ≥ 200 mg/dL (11.1 mmol/L)**
- **WHO(2011) : HbA1c >6.5% (48 mmol/mol)**

Prediabetic > 100 mg/dL

CLASSIFICATION OF DIABETES MELLITUS

- **Insulin dependent Diabetes mellitus (IDDM, Type 1, juvenile onset)**
  - insulin deficiency
  - acute, severe
  - diabetic ketoacidosis (DKA)
  - autoimmune disease
  - ~10%

- **Non insulin dependent Diabetes mellitus (NIDDM, Type 2, maturity onset)**
  - insulin resistance
  - relative insulin deficiency
  - mild to moderate
  - related to obesity
  - complication
    (microvascular : retina, renal, brain
    macrovascular : atherosclerosis)
  - hyperosmolar coma

Diabetic Treatment Goals

รักษาระดับน้ำตาลในเลือดให้อยู่ในระดับปกติ

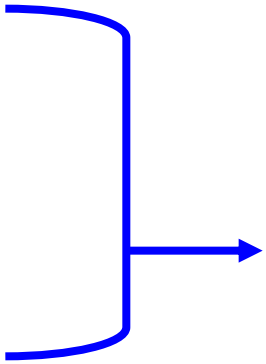
- **Fasting blood glucose : 70-99 mg/dL(3.9-5.5 mmol/L)**
- **2 hours after eating : 70-145 mg/dL (3.9-8.1 mmol/L) (postprandial)**
- **Random (casual) : 70-125 mg/dL (3.9-6.9 mmol/L)**

- ***Reducing hyperglycemia ⇒ reduces the risk of microvascular & macrovascular complications of diabetes (retina, kidney, nerve, heart)***

Diabetes Treatment Guidelines

- Diet-controlled

- Excercise



Dietary Guidelines :

- plant-based
- high in fiber
- low in animal fat
- 5-9 servings of fruits and vegetables/day

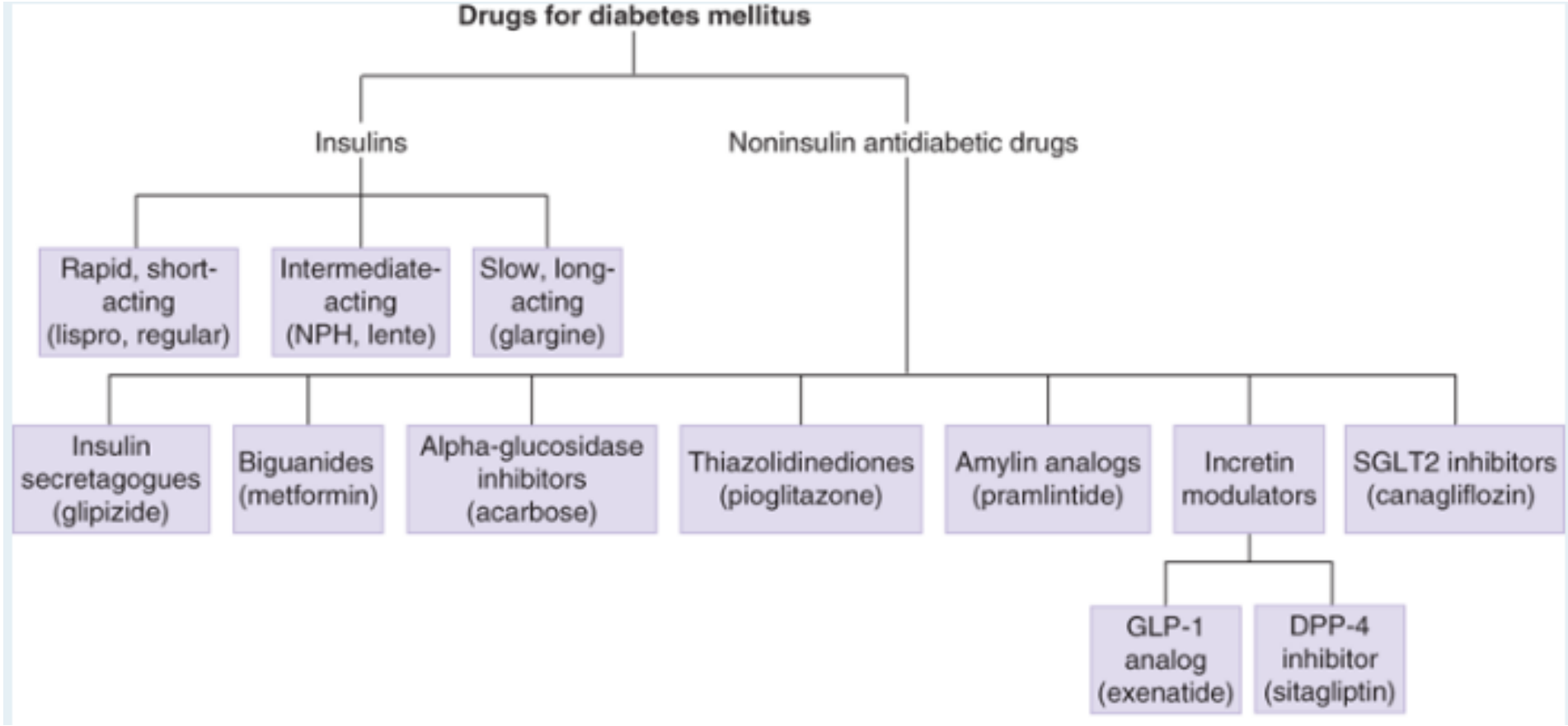
Improve insulin resistance, hyperglycemia, hypertension, lipid profile

- Hypoglycemic drugs

HYPOGLYCEMIC DRUGS

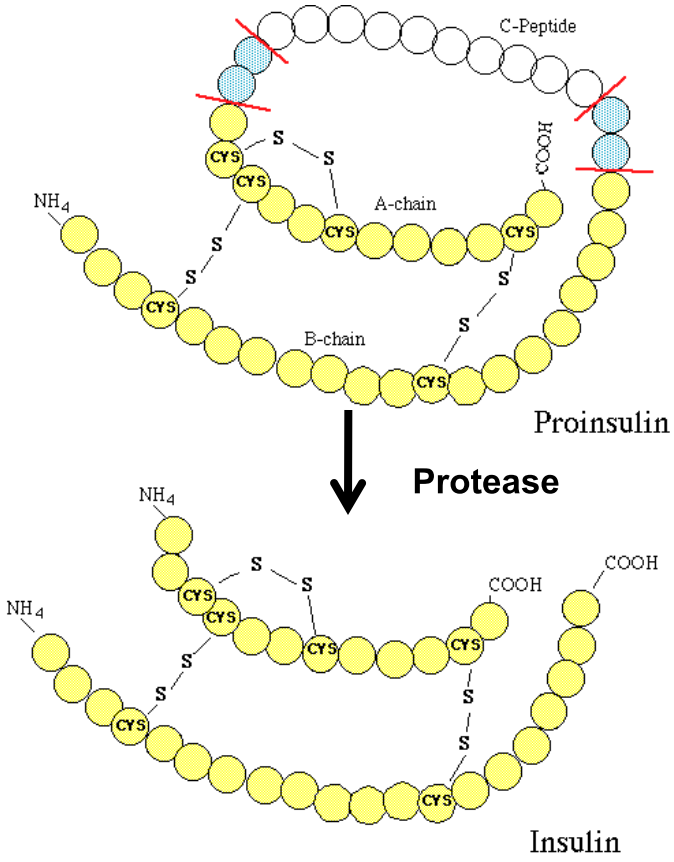
- **Injectable drugs**
  - Insulin
  - Incretin mimetics (GLP-1)
  - Amylin analogs
- **Oral hypoglycemic drugs**
  - Insulin secretagogues
  - Insulin sensitizers
  - Alpha-glucosidase inhibitors
  - DPP-4 inhibitors
  - Sodium-glucose transporter inhibitors
  - Centrally Acting Drug
  - Dual-acting glucokinase activator

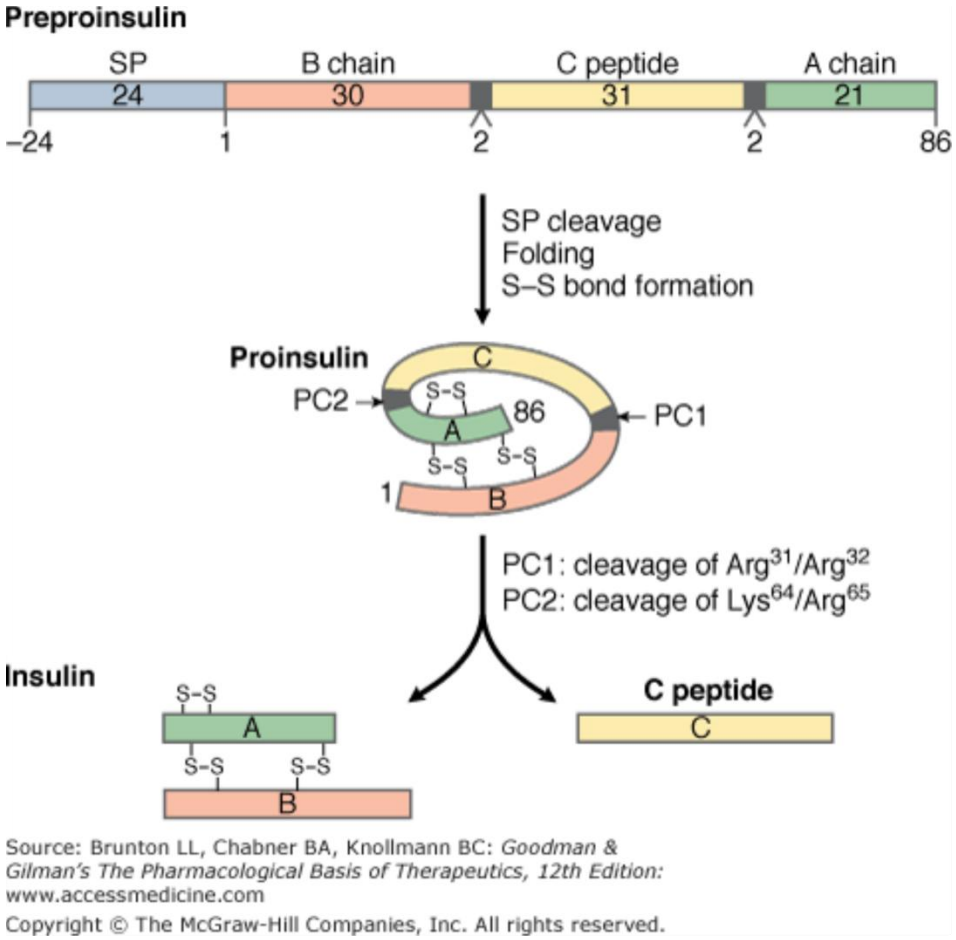
Antidiabetic drugs



INSULIN

Structure : polypeptide hormone
Source : recombinant DNA technology (human insulin)

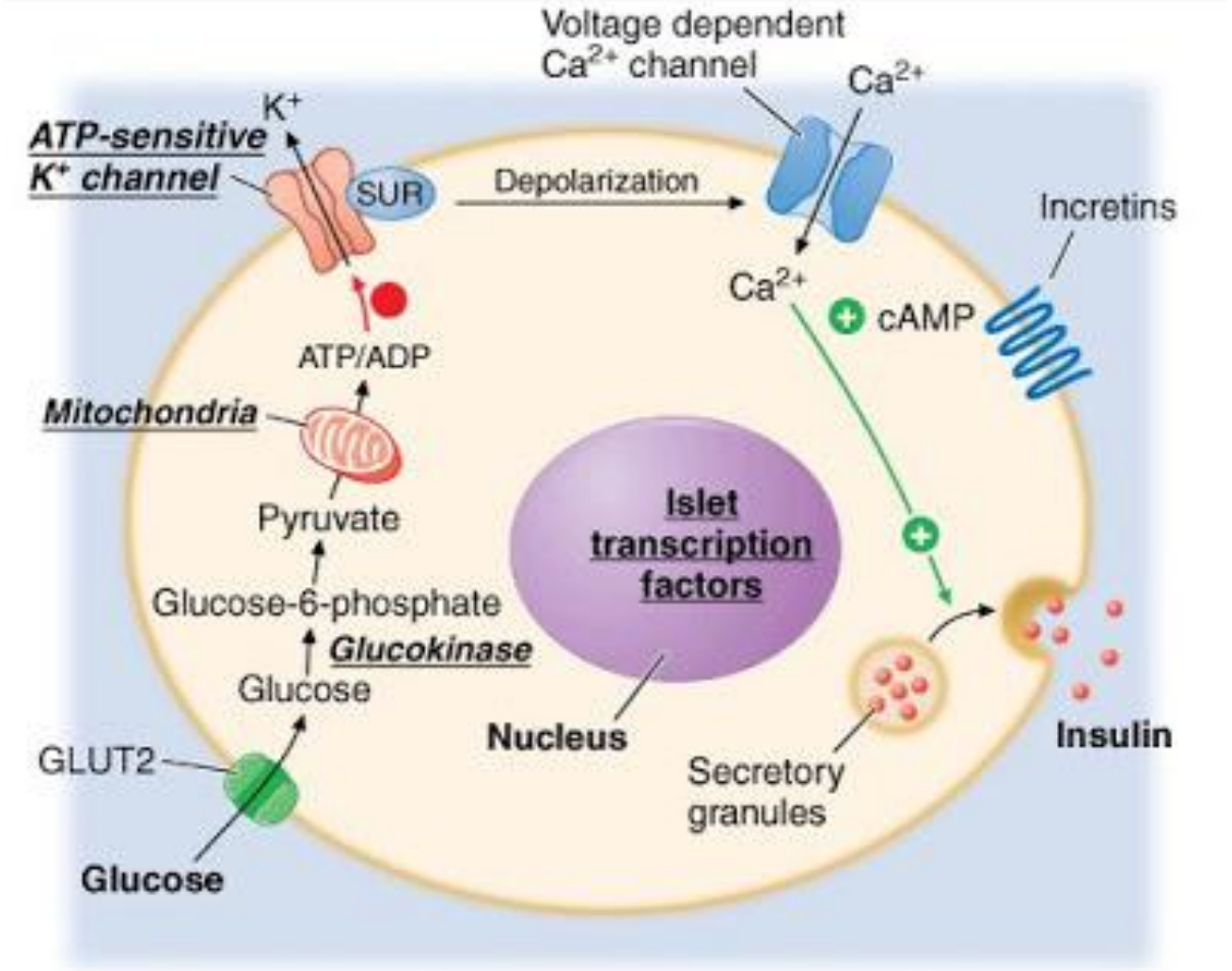




Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com

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Regulation of insulin secretion



Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 17th Edition: http://www.accessmedicine.com

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The insulin receptor

- Insulin receptor: $\alpha_2\beta_2$
  - α extracellular, binds insulin
  - β transmembrane domains and tyrosine kinase
  - Autophosphorylation and phosphorylation of other proteins

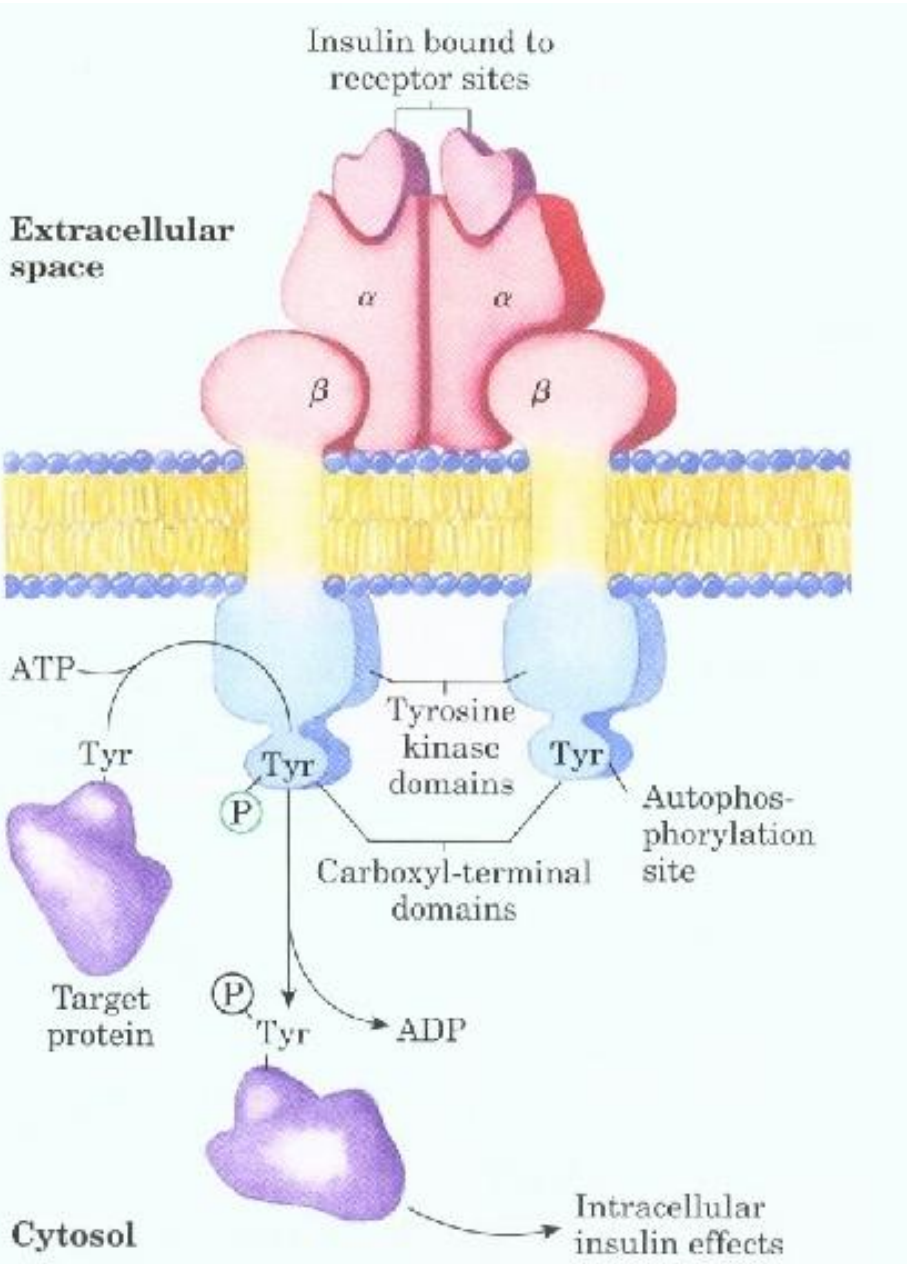
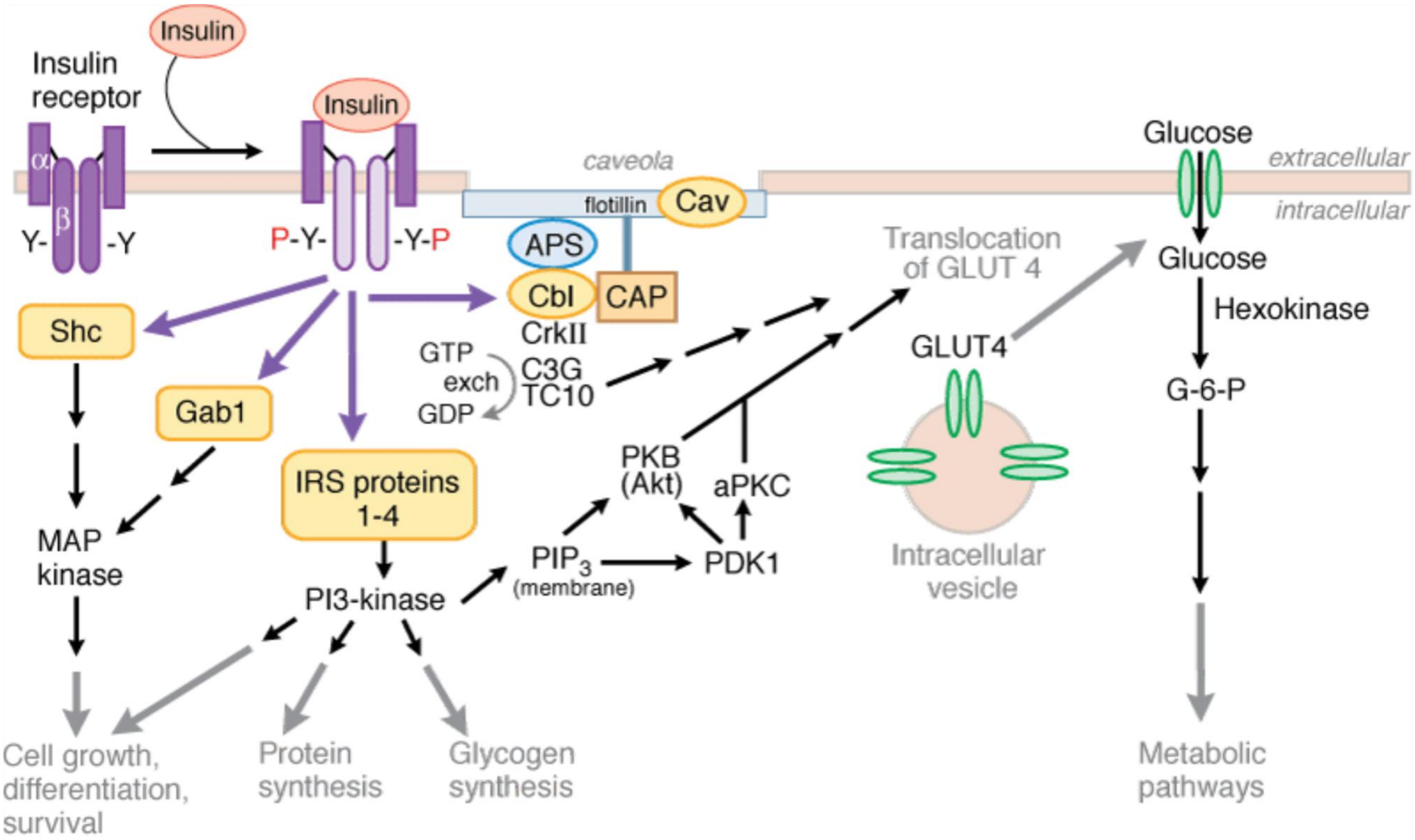


figure 13-6 Lehninger Principles Bchm 2000

Insulin signaling pathway

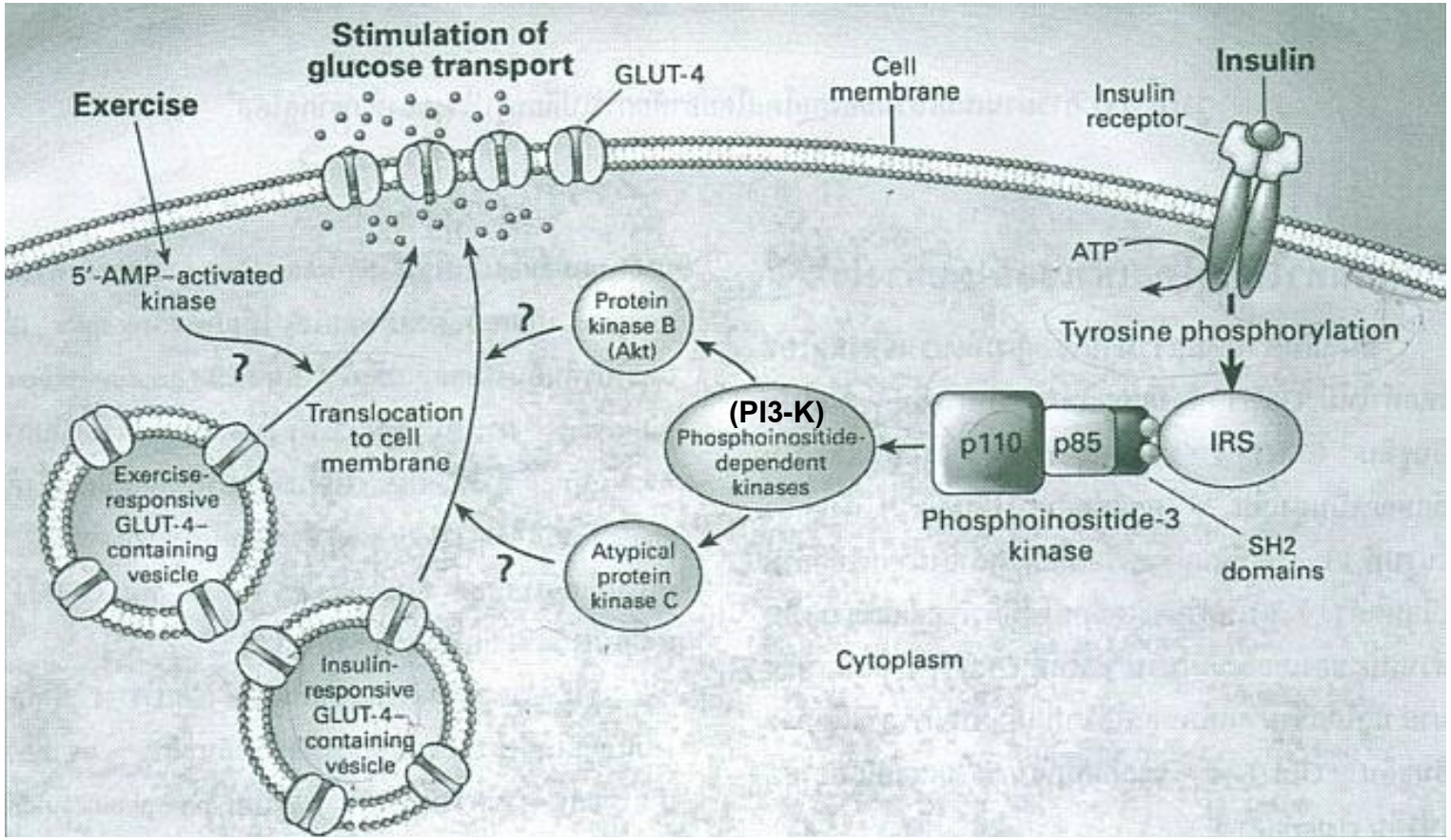


Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com

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PHARMACOLOGICAL EFFECTS

- **Effect on glucose transporter : GLUT4, GLUT2**



- **Effect on liver**

  - Reversal of catabolic features of insulin deficiency

  : inhibits glycogenolysis

  : inhibit conversion of fatty acids and amino acids to keto acids

  : inhibit conversion of amino acids to glucose

  - Anabolic action

  : promotes glucose storage as glycogen

  (induce glucokinase & glycogen synthase, inhibit phosphorylase)

- **Effect on muscle**
  - ↑ protein synthesis
    - : ↑ amino acid transport
    - : ↑ ribosomal protein synthesis
  - increase glycogen synthesis
    - : ↑ glucose transport
    - : induce glycogen synthase
    - : inhibits phosphorylase

- **Effect on adipose tissue**

  - ↑ triglyceride storage

  : induce lipoprotein lipase to hydrolyze triglycerides from lipoproteins

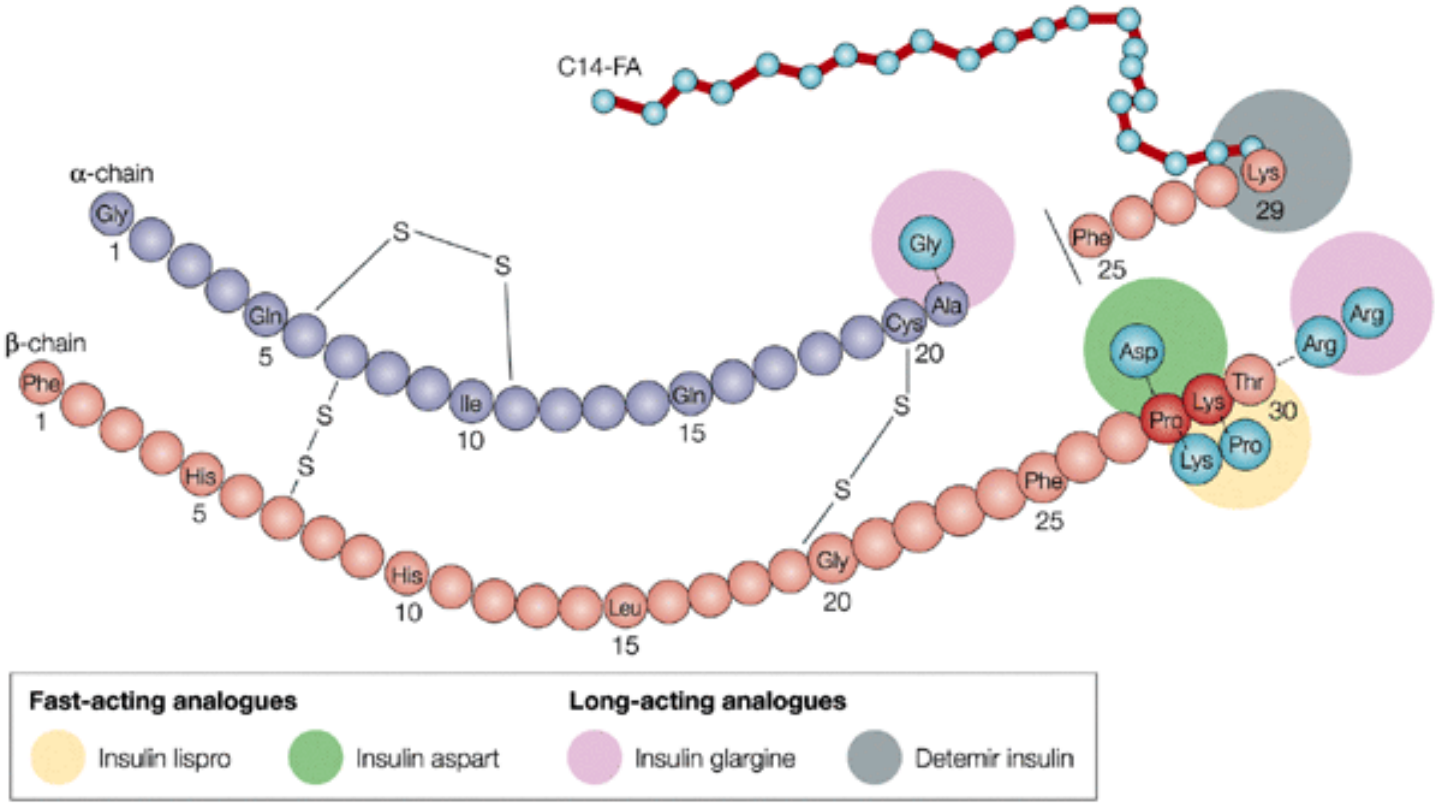
  : ↑ glucose transport into cell provides glycerol phosphate to permit esterification of fatty acids

  : inhibit intracellular lipase, reduce lipolysis

INSULIN PREPARATIONS

- **Ultrashort-acting :** ***Insulin lispro, Insulin aspart***
- **Short-acting :** ***Regular insulin (RI)***
- **Intermediate-acting :** ***Lente insulin (+Zn)***

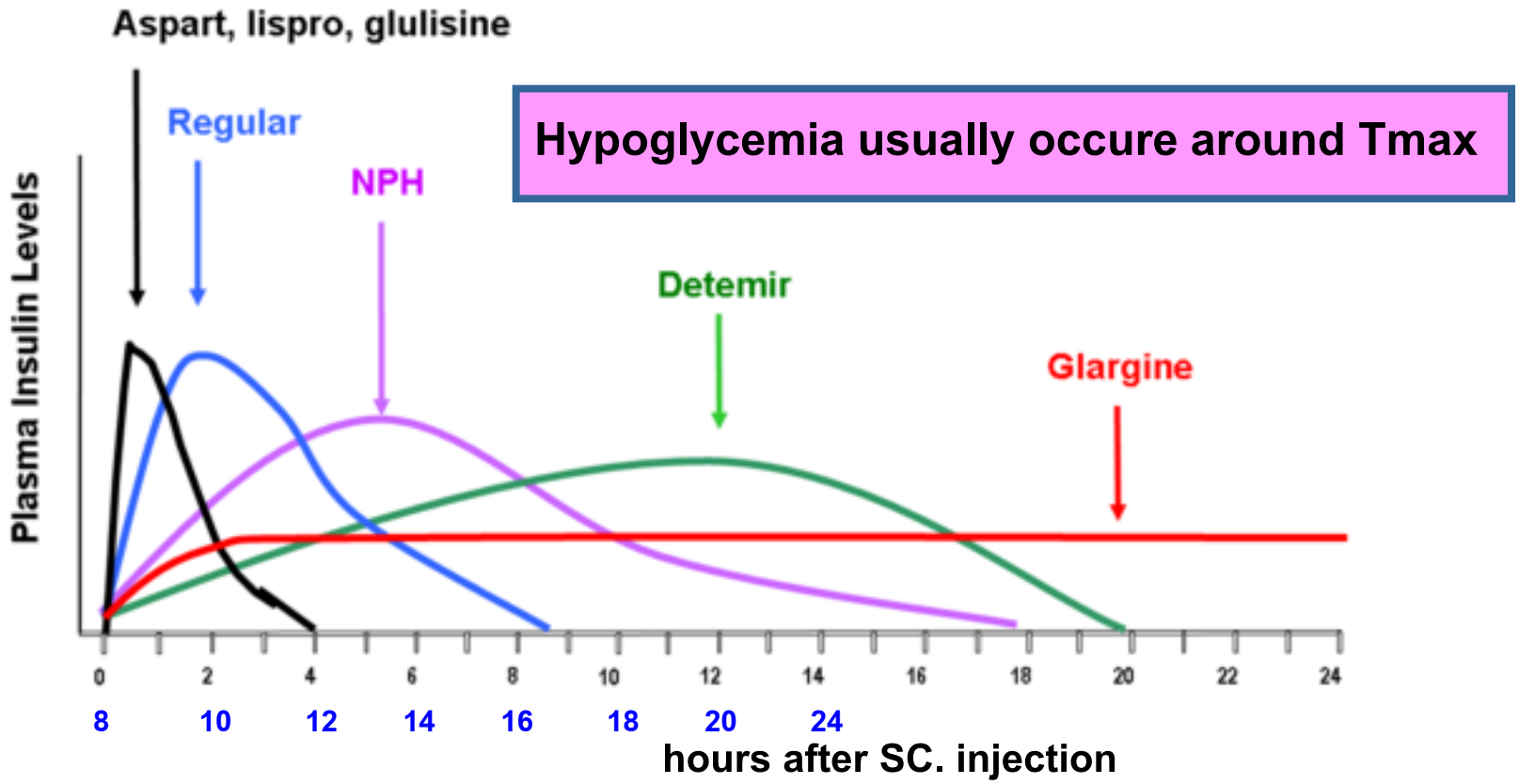
  : ***Neutral Protamine Hagedron (NPH, Isophane insulin)***
- **Long-acting :** ***Insulin glargine, Ultralente, Protamine Zinc insulin (PZI)***
- **Premixed insulin :** ***short acting + intermediate-acting***



Nature Reviews | Drug Discovery

- **Insulin lispro** creates steric hindrance and a reduced ability to self-associate
- **Insulin aspart** create charge repulsion and steric hindrance
- **Insulin glargine** contains two extra arginine to alter isoelectric point and reduce its solubility, a glycine substitution was made to stabilize the molecule
- **Insulin detemir**, acetylation of Lys B29, promote reversible binding of insulin to albumin and delay the absorption

Type of insulin	Onset (hr)	Duration (hr)	Peak (hr)
Ultrashort-acting			
- Insulin lispro	0.25	5	0.5-1.5
Short-acting			
- Regular insulin (RI)	0.5-1	4-6	2-4
Intermediate-acting			
- NPH	1-4	12-20	4-10
- Lente insulin	2-4	12-20	8-12
Long-acting			
- Glargine	3-5	18-24	(10-16)



Hypoglycemia usually occure around Tmax

INDICATION

- Type 1 diabetes
- Uncontrolled Type 2 diabetes
- Pregnancy, Stress (operation, infection, myocardial infarction)
- Hyperkalemia (+ glucose)

ROUTES OF ADMINISTRATION

- IV for diabetic ketoacidosis (RI)
- SC for routine use
- Inhalation (2006-2008)
- Patch (2010)





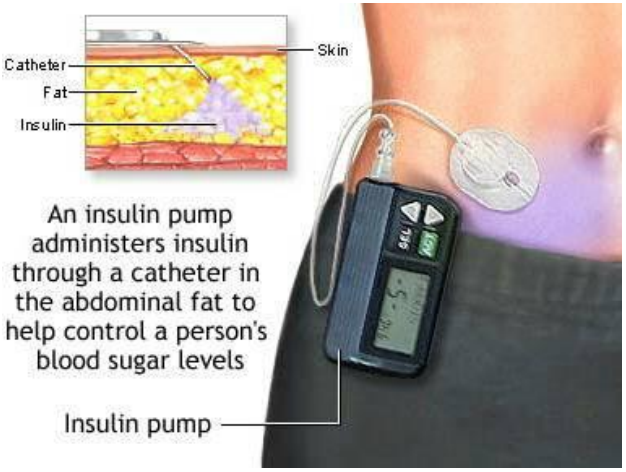


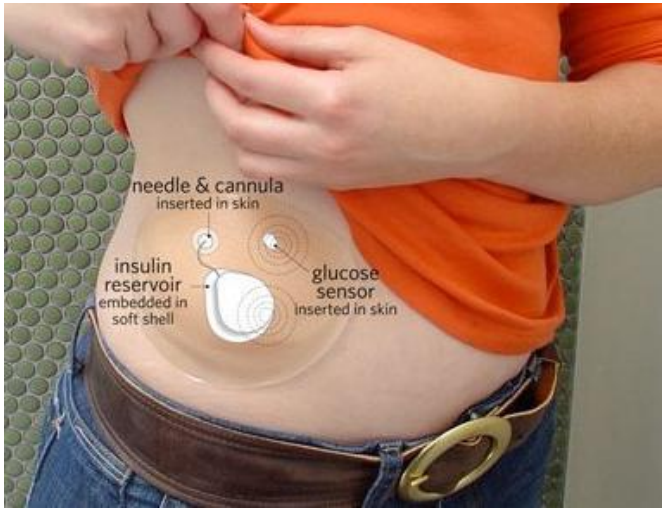


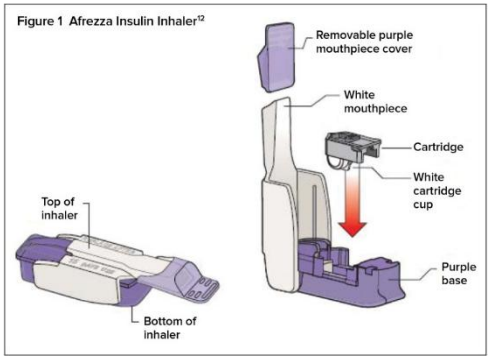


(exubera® approved 2006, withdrawn 2008)









On June 27, 2014,

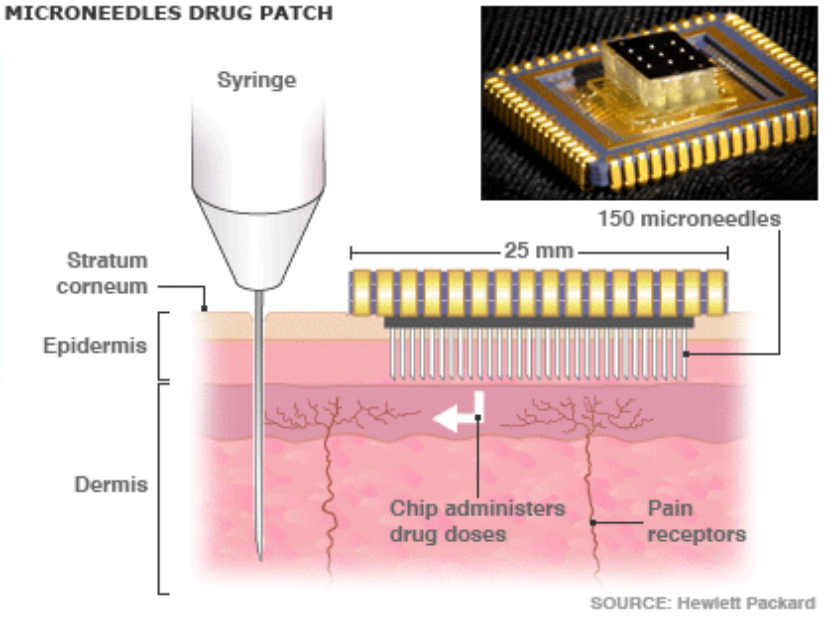
the FDA approved Afrezza



Ultrasonic transdermal patch



Insulin patch



SIDE EFFECTS

- Hypoglycemia
- Lipodystrophy : *hypertrophy*

  : *atrophy*
- Obesity, edema (Na^+ retension, ↑vascular permeability)
- Allergy (rare) : immediate type I hypersensitivity
- Insulin resistance : IgG anti-insulin antibody

HYPOGLYCEMIA

- **Causes:** delay in eating, unusaul physical exertion, inappropriate high dose
- Plasma glucose ~ 70mg/dl (3.9 mM)

- release of counterregulatory hormone
  : epinephrine, norepinephrine, glucagon, growth hormone, cortisol
- symptoms : sweating, tremor, palpitation, anxiety hunger, nausea, paresthesia
- treatment : sugar containing beverage

- **Lower plasma level ~ 40 mg/dl**

  - neuroglycopenic symptom : confusion, weakness, drowsiness, blurred vision, loss of conciousness, *convulsion, coma*

  - treatment : IV glucose, IM glucagon

LIPODYSTROPHY

- Atrophy

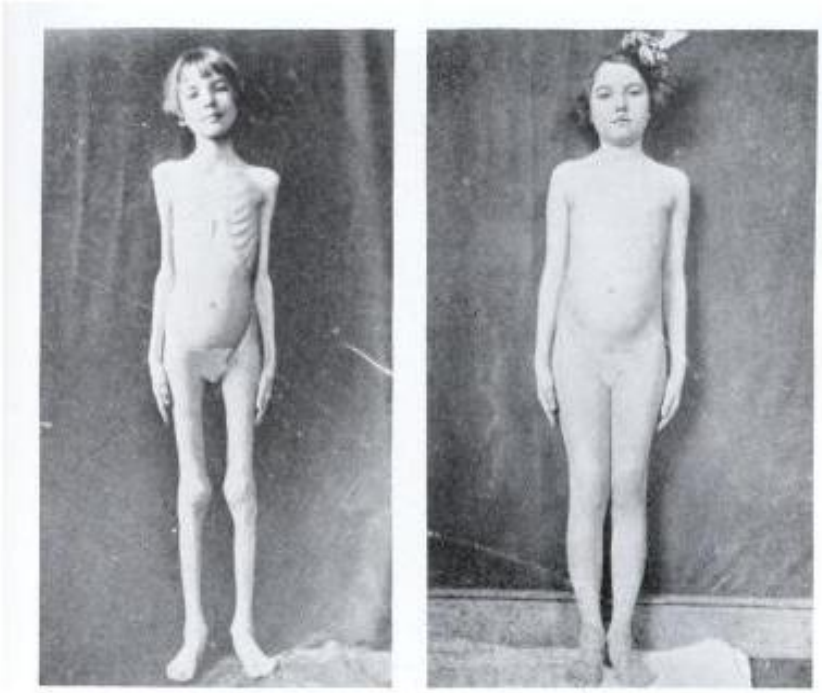






- Hyperdystrophy





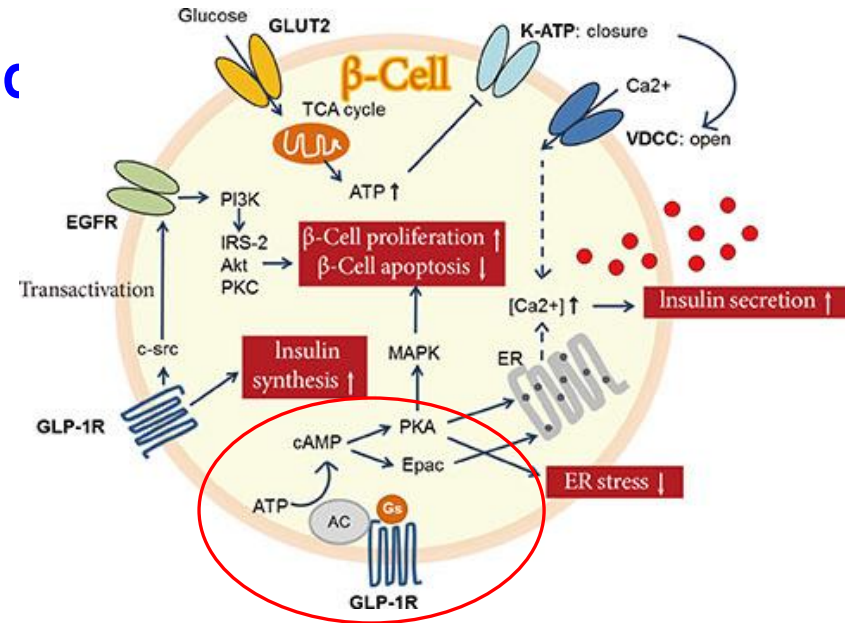
- A girl with diabetes before all insulin treatments and after beginning treatments.
- These pictures were taken in 1922, soon after the discovery of insulin

  (Bliss, M. 1982. The Discovery of Insulin. Chicago: The University of Chicago Press.

Incretin mimetics (glucagon-like peptide-1, GLP-1)



- **Exenatide, Liraglutide** (peptide originally isolated from the Gila monster saliva) : approved for Type 2 DM by FDA in 2005
- Lixisenatide, Dulaglutide, Albiglutide, Liraglutide
- **Action** : - Stimulate glucose-dependent insulin secretion
  - Reduce glucagon secretic
  - Slow gastric emptying



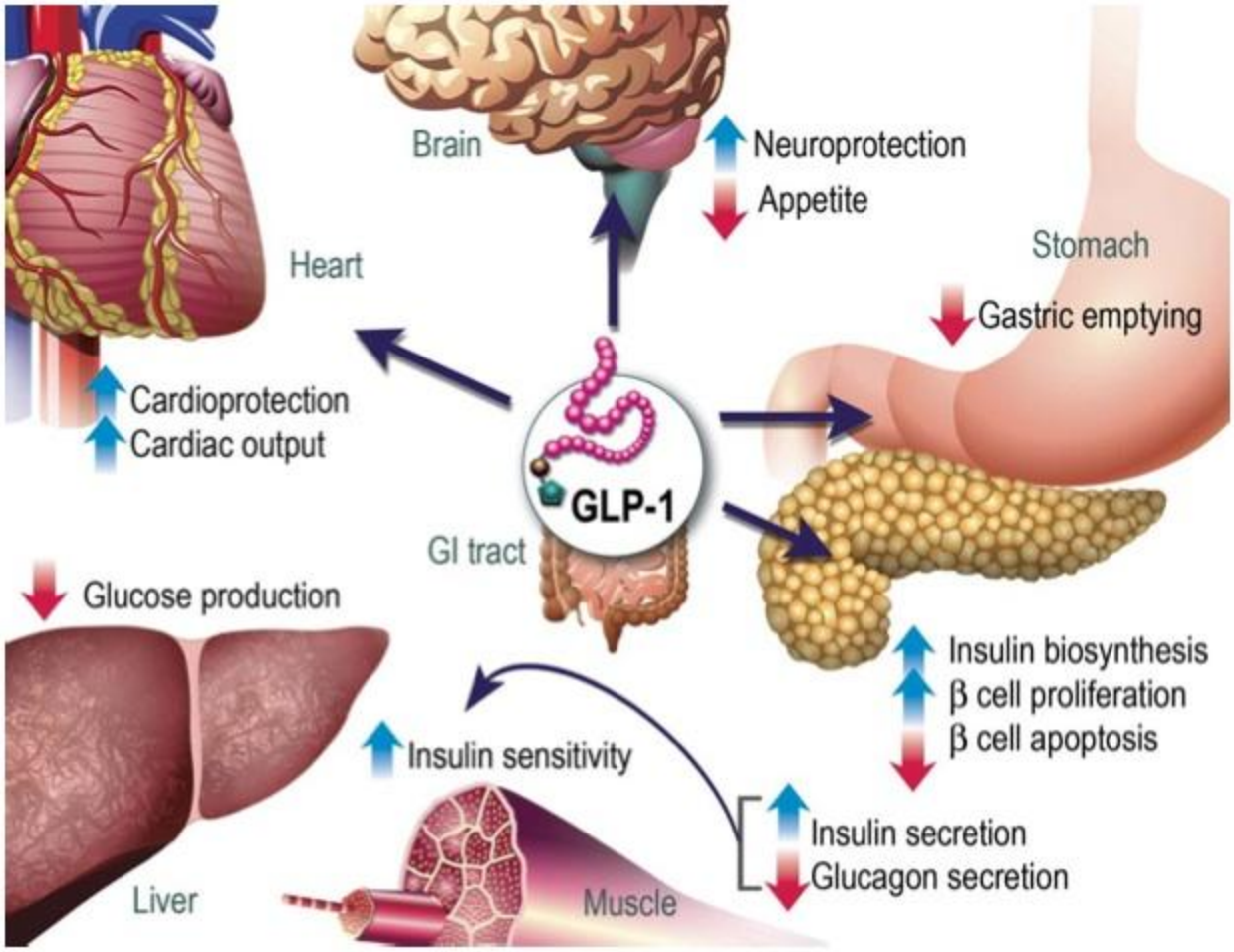


Figure 1. GLP-1 actions in peripheral tissues

GLP-1 acts directly on the endocrine pancreas, heart, stomach, and brain, whereas actions on liver and muscle are indirect.

- **Excretion : excreted by glomerular filtration**
- **Side effects : nausea, diarrhea, dyspepsia, dizziness, pancreatitis?**
- **Indication : add-on therapy with oral hypoglycemic drugs**
- **Contraindication : hypersensitivity, Type 1 DM, renal & GI disease**
- **Route : SC**

Amylin analog

- Amylin : a peptide neurohormone co-secreted with insulin from beta cell
- Pramlintide (Symlin) : amylin-mimetic
- Action :

  - slow gastric emptying $\Rightarrow$ early satiety, $\downarrow$food intake, weight loss
  - $\downarrow$ glucagon release in post prandial state
  - $\downarrow$ hepatic glucose production
- Indication : T1DM, T2DM, SC, before each meal
- Side effects : hypoglycemia, nausea, vomiting, abdominal pain

ORAL HYPOGLYCEMIC DRUGS

- Insulin secretagogues : sulfonylureas, nateglinide, meglitinide
- Insulin sensitizers : biguanide, glitazones
- alpha-glucosidase inhibitors : acarbose, voglibose, miglitol
- incretin mimetics, DPP4-inhibitors : sitagliptin, vildagliptin, alogliptin
- Sodium-glucose transport protein (SGLT2) inhibitor : dapagliflozin, canagliflozin, empagliflozin

INSULIN SECRETAGOGUES

Promote insulin release from β-cells of pancreas

- **Sulfonylureas**

  - *first generation : tolbutamide, chlorpropamide*

  - second generation

  : glipizide, glicazide, glimepiride (short acting)

  : glibenclamide (glyburide) (long- acting)

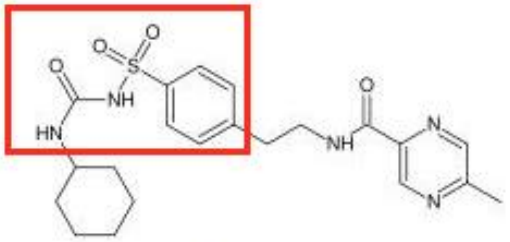
- **Meglitinide analogs : repaglinide**

  (rapid onset & short acting)

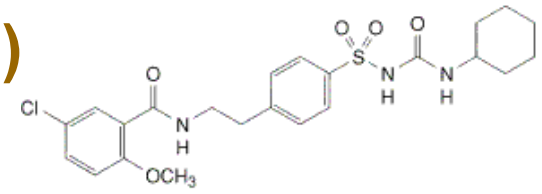
- **D-Phenylalanine derivative : nateglinide**

  : meglitinide

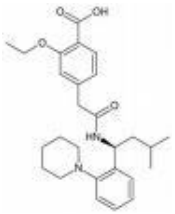
sulfonyl urea

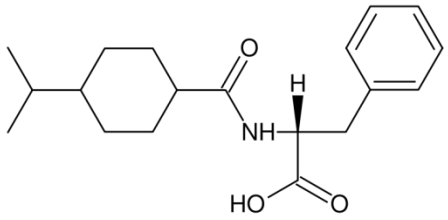


Glipizide



glibenclamide

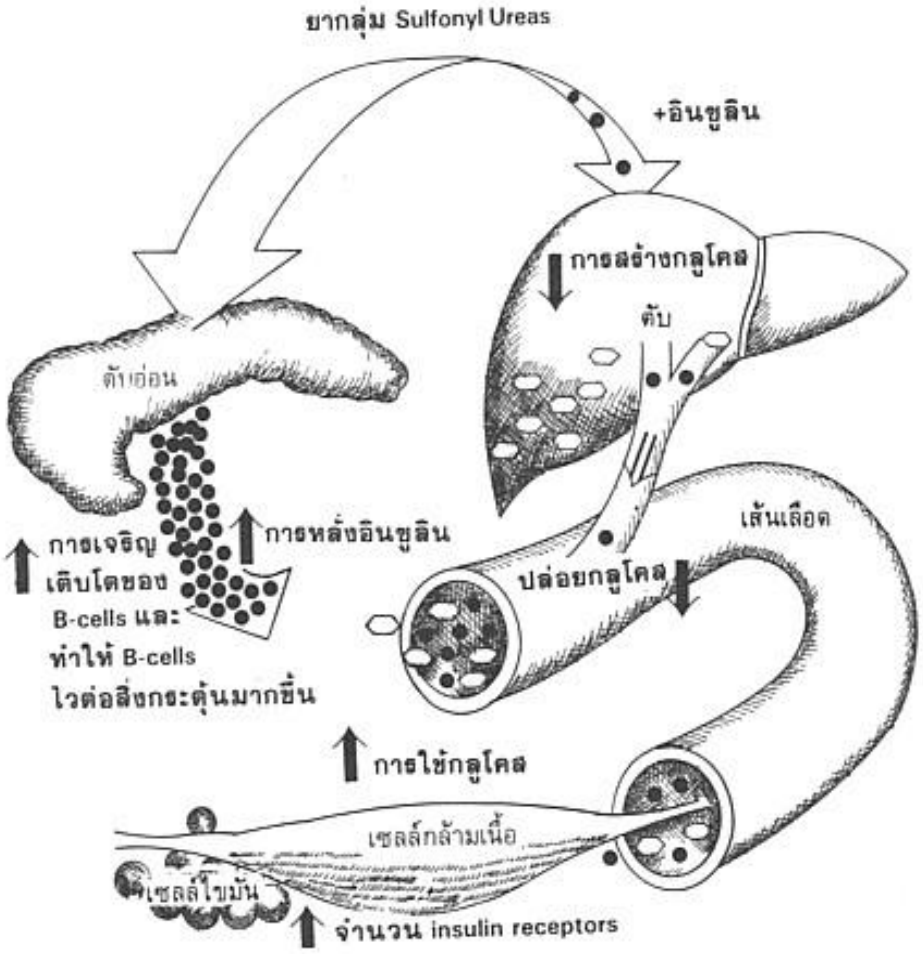




ACTIONS OF SULFONYLUREAS

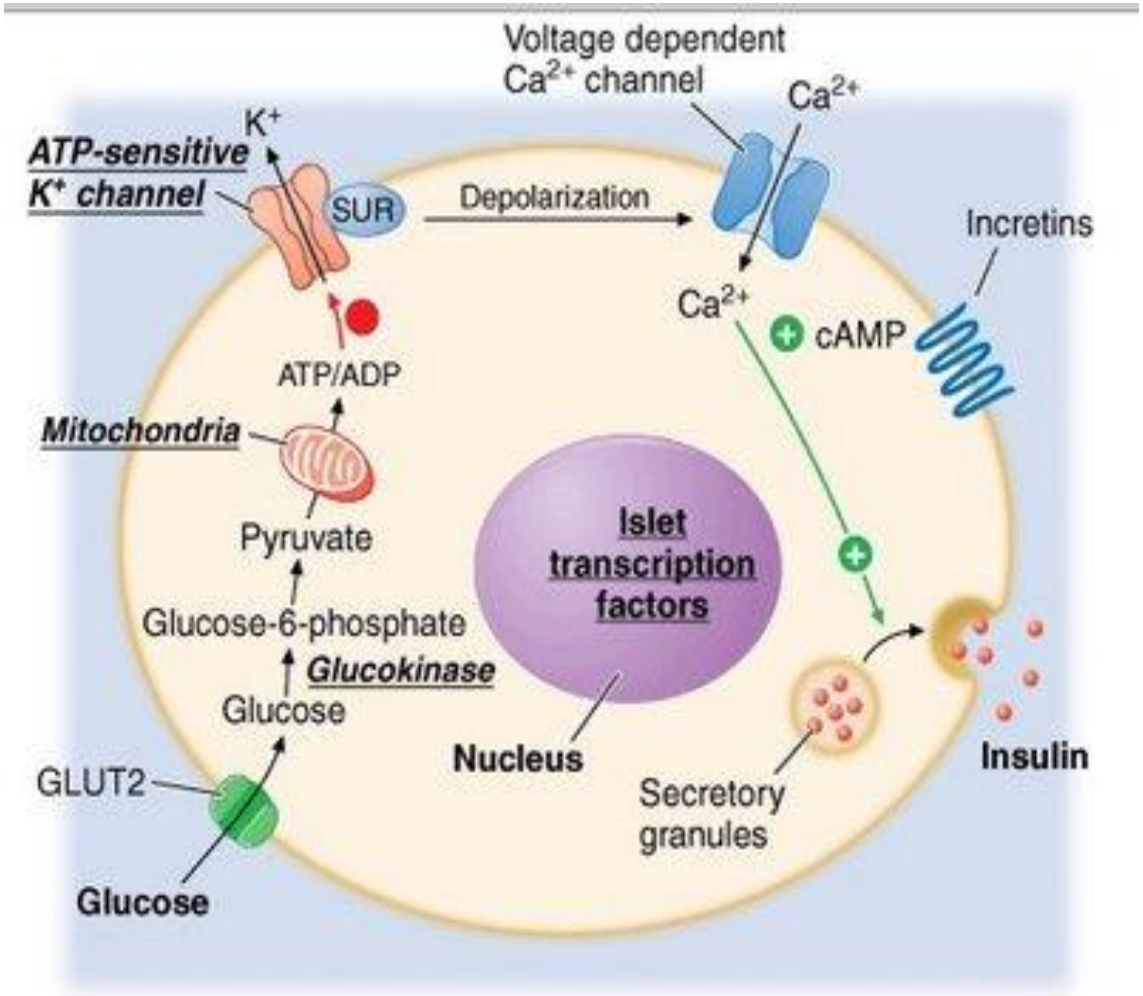
- ↑ insulin secretion
- ↓ hepatic glucose output
- ↓ serum glucagon
- ↑ insulin sensitivity

  (↑affinity, insulin receptor)



MECHANISM OF ACTION

- Block ATP-sensitive K^+ channel



SIDE EFFECTS

- Hypoglycemia – long acting, elderly, renal & hepatic complication
- Weight gain
- Water retention : edema
- Disulfuram-like effect (rare)

DRUG INTERACTION

- ↑effect : sulfonamides, salicylates, dicumarol, clofibrate, alcohol
- ↓ effect : epinephrine, corticosteroid, thiazide diuretics
- ↓ sympathetic activity : propranolol

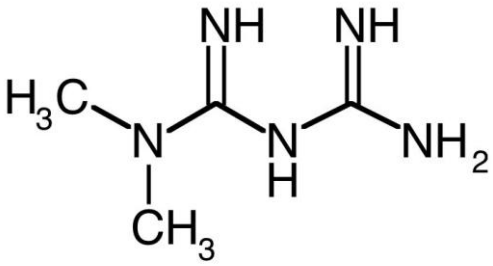
INSULIN SENSITIZERS

- **BIGUANIDES : metformin**
- **THIAZOLIDINEDIONE (glitazone) - pioglitazone**

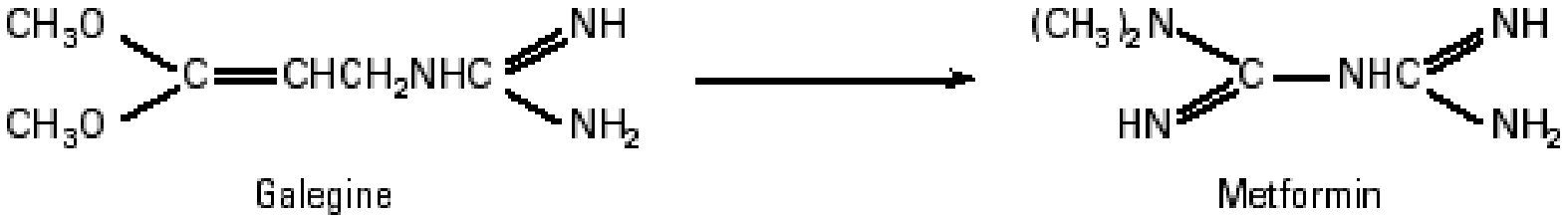
BIGUANIDES
(metformin)

(synthesis 1920, clinical use 1957, FDA approved 1994)





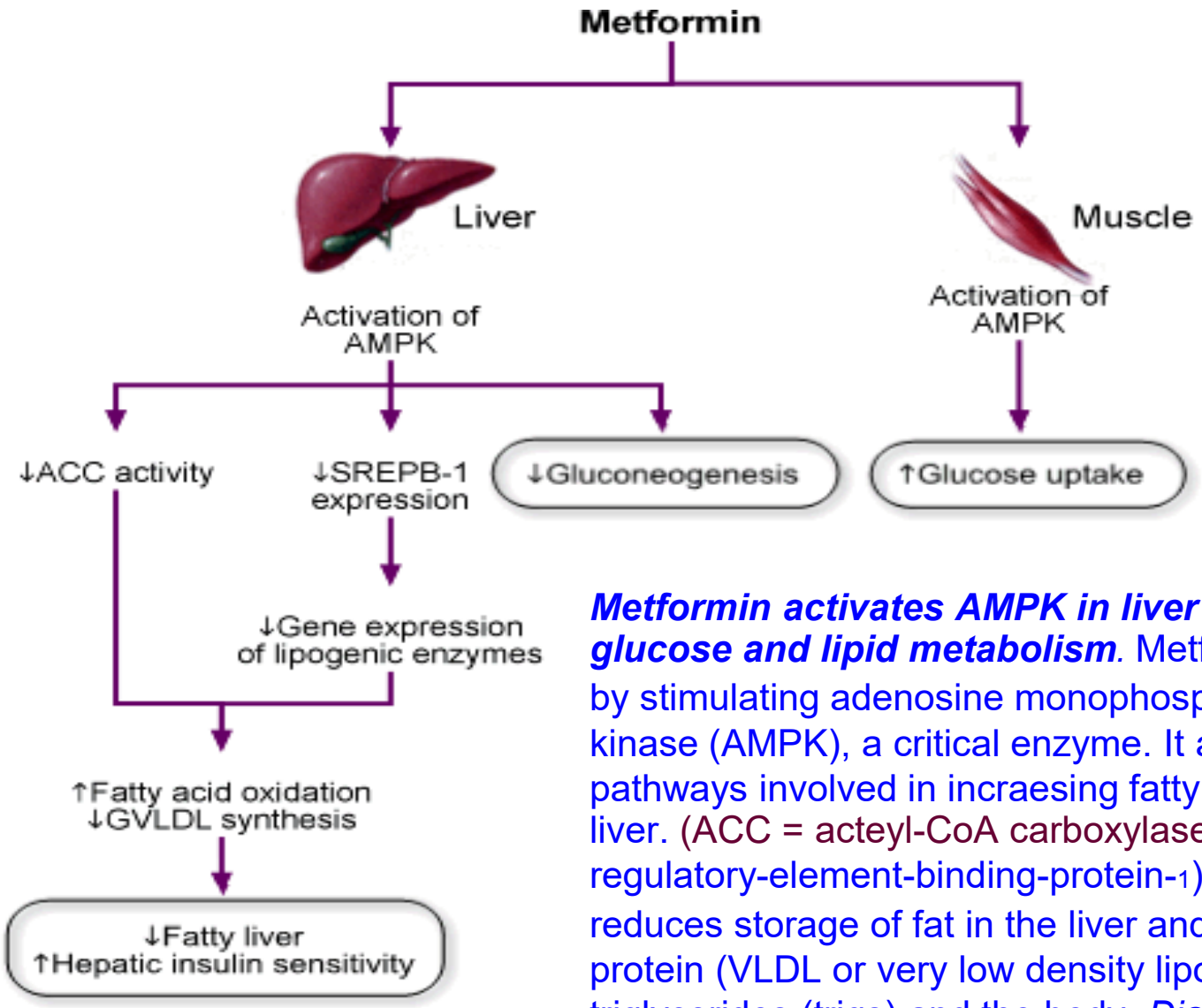
***Galega officinalis*, a natural source of galegine**



Others: buformin

- **ACTIONS**
  - ↓ hepatic gluconeogenesis
  - ↓ glucose absorption from GI tract
  - ↑ glucose uptake into skeletal muscle
  - ↑ glucose removal from blood : stimulate anaerobic glycolysis

    ⇒ lactic acid
  - ↓ LDL, triglyceride



Metformin activates AMPK in liver and muscle to improve glucose and lipid metabolism. Metformin mediates its action by stimulating adenosine monophosphate-activated protein kinase (AMPK), a critical enzyme. It also reduces enzymatic pathways involved in incraesing fatty acid production by the liver. (ACC = acteyl-CoA carboxylase; SREPB-1 = sterol-regulatory-element-binding-protein-1) In this manner it reduces storage of fat in the liver and in the blood carrier protein (VLDL or very low density lipoprotein) that shuttles triglycerides (trigs) and the body. *Diagram adapted from Alice Y.Y. Cheng, I. George Fantus, 'Oral antihyperglycemic therapy for type 2 diabetes mellitus' Canadian Medical Association Journal 172(2), 2005 pp213-226.*

Clinical pharmacology

In overweight people, >10 years of treatment

- diabetes complications (heart attacks, strokes and other cardiovascular complications) and overall mortality decreased by metformin
  - > insulin & glibenclamide & chlorpropamide = 30%
  - > dietary advice = 40%
- Weight gain : metformin, dietary advice 1 kg
  : sulfonylurea 3 kg, insulin 6 kg
- Blood sugar control : metformin ~ sulfonylurea, insulin
- Risk of hypoglycemia : metformin < sulfonylureas

SIDE EFFECTS

- GI disturbance

  : anorexia, nausea, vomiting, diarrhea, weight loss (dose-related, usually occurs at the initiation, often transient)
- Vitamin B12 Deficiency (long term treatment)
- Lactic acidosis : fatal ! (mortality rate 50%), rare (5.1/100,000)
- Alopecia (0.6%)
- Erosive lichen planus at buccal mucosa

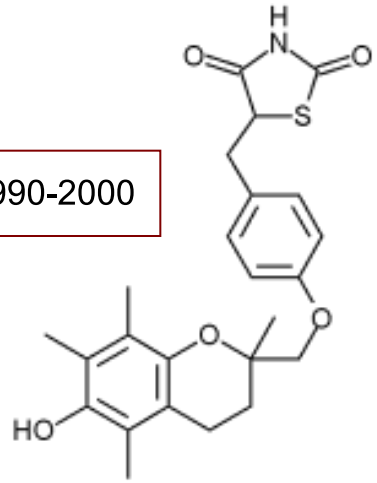
CONTRAINDICATION

- Renal disease (creatinine levels over 1.5-1.7 mg/dL)
- Alcoholism
- Hepatic disease
- Chronic cardiopulmonary disease : heart failure (associated with tissue hypoxia)
- Severe infection
- Age 80+

THIAZOLIDINEDIONE
(glitazones)

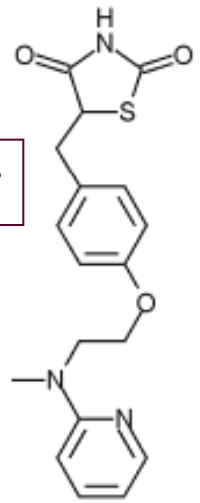
troglitazone

1990-2000

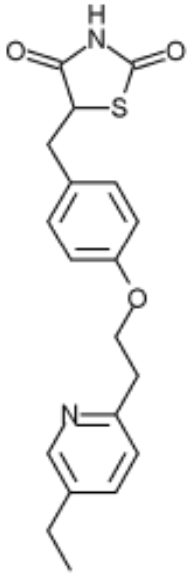


rosiglitazone

1999-2011?



pioglitazone

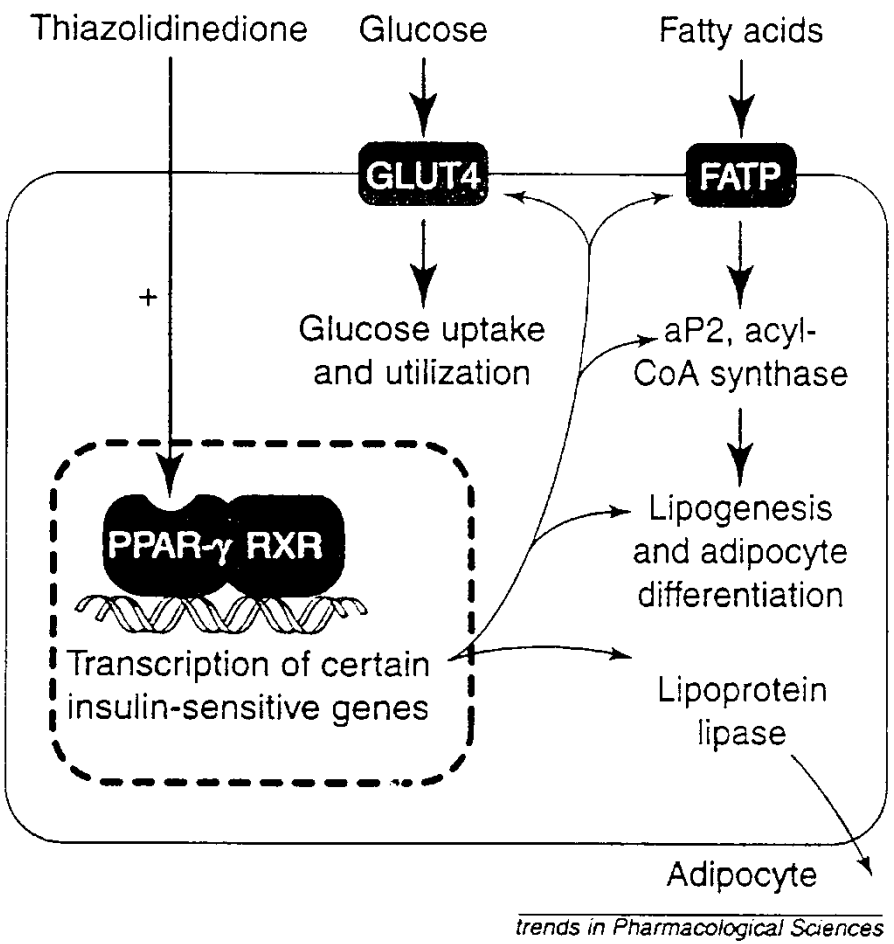


ACTIONS OF THIAZOLIDINEDIONE

- ↑ insulin sensitivity
- ↑ glucose uptake
- ↓ hepatic gluconeogenesis

Mechanism of action

- Control transcription of insulin-sensitive gene
- ↓ Insulin resistance
- ↑adipose cell differentiation
- ↑adiponectin
- ↓ leptin, resistin, TNFα, IL-6



PPAR-γ = peroxisome proliferator-activated receptor gamma

SIDE EFFECTS

- Fluid retention *(promote amiloride-sensitive sodium ion reabsorption in renal collecting duct, increase plasma volume 500 ml)*, edema, congestive heart failure
- Mild anemia *(haemodilution, bone marrow)*
- Weight gain (1-4 kg)
- Mild increase LDL & HDL cholesterol
- Increase deposition of subcutaneous fat
- Osteoporosis
- Hepatotoxicity !
- Pale

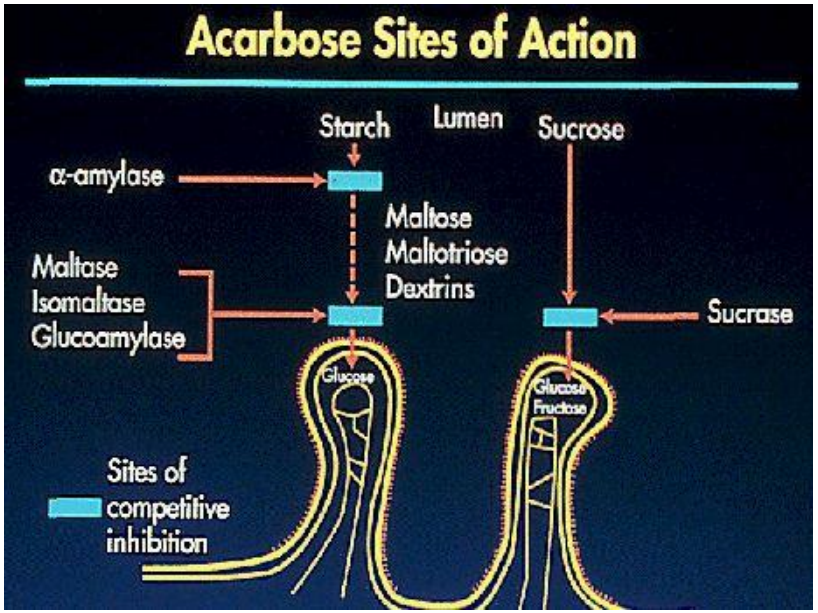
- [**2-03-2011**] The U.S. FDA is notifying the public that information on the cardiovascular risks (including heart attack) of the diabetes drug rosiglitazone has been added to the physician labeling and patient Medication Guide.
- **ยารักษาโรคเบาหวาน ยี่ห้อ "อาแวนเดีย (Avandia)" ตัวยาสำคัญคือยาโรซิกลิทาโซน (Rosiglitazone) อย.ได้มีการเพิกถอนทะเบียนตำรับยาไปตั้งแต่ต้นปี 2554 เนื่องจากมีข้อมูลความเสี่ยงต่อระบบหัวใจและหลอดเลือด.**
- [**6-15-2011**] The U.S. Food and Drug Administration (FDA) is informing the public that use of the diabetes medication Actos (pioglitazone) for more than one year may be associated with an increased risk of bladder cancer

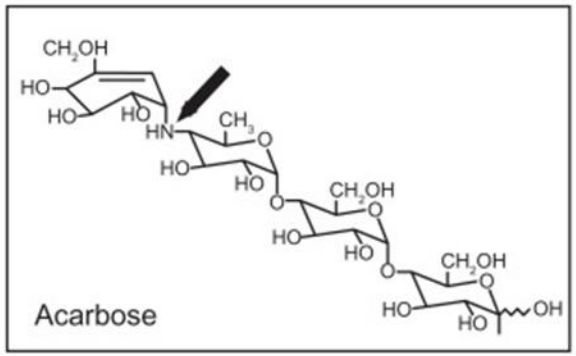
CONTRAINDICATION

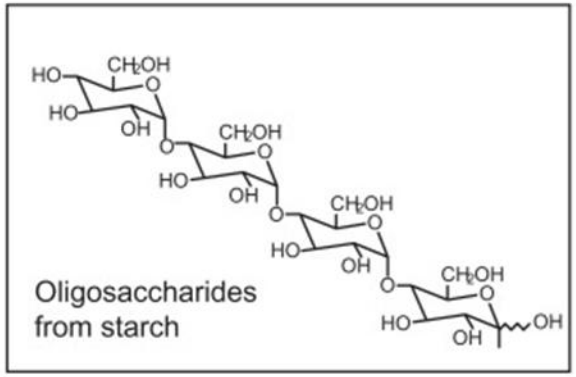
- Pregnancy
- liver disease
- heart failure

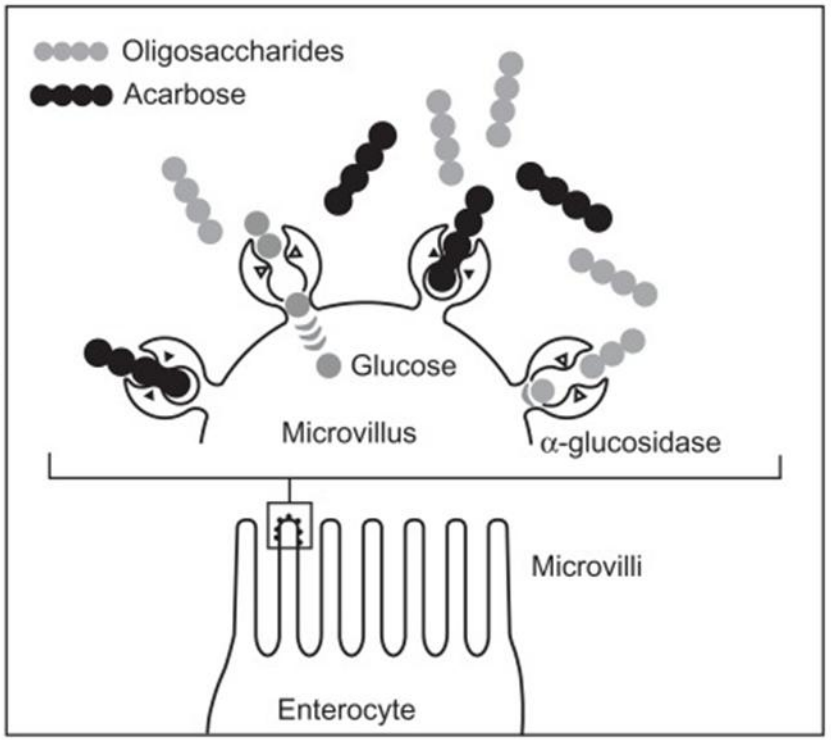
Alpha-glucosidase inhibitors

: acarbose, voglibose, miglitol









- **Mechanism of action**

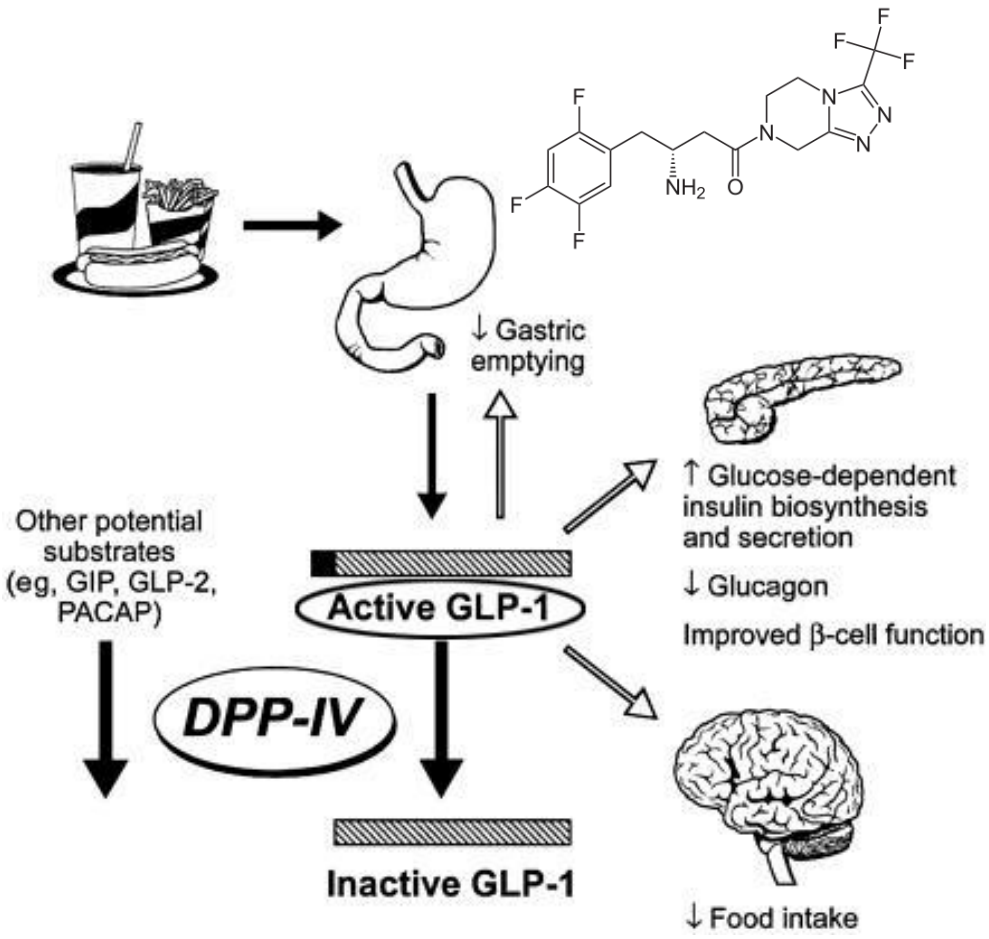
  : strongly bind to intestinal disaccharidases *e.g.*

  alpha- glucosidase **$\Rightarrow$ delayed carbohydrate absorption**

- **Side effects : flatulence, diarrhea, hepatotoxicity**
- **Contraindication : inflammatory bowel disease, renal disease**

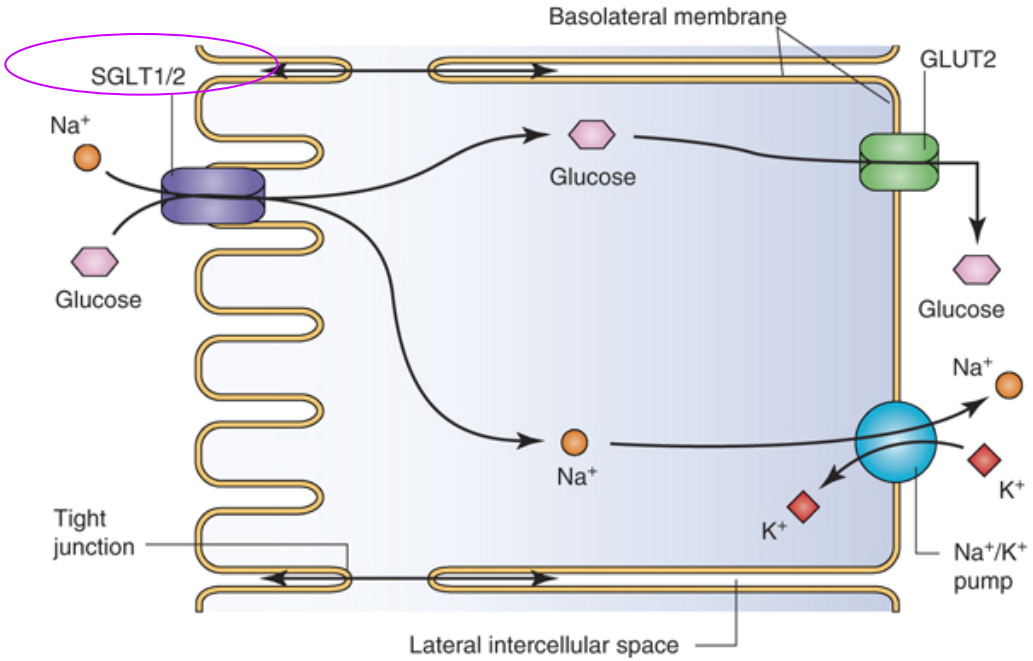
Dipeptidyl Peptidase-4 (DPP-4) inhibitors

- **Sitagliptin : Thai FDA approved 2551 (@200-250 baht)**
- **Others: saxagliptin, linagliptin**
- **Indication : once daily, orally as monotherapy or combination for T2DM**
- **Side effects : diarrhea, nausea, abdominal pain, hypoglycemia, pancreatitis?**



Sodium-glucose transport protein (SGLT2) inhibitor

- Dapagliflozin (Marketed in EU 2012)
- Others : canagliflozin, empagliflozin
- SGLT2: reabsorb filtered glucose in tubular lumen
- Potential side effects : urinary tract & genital infections



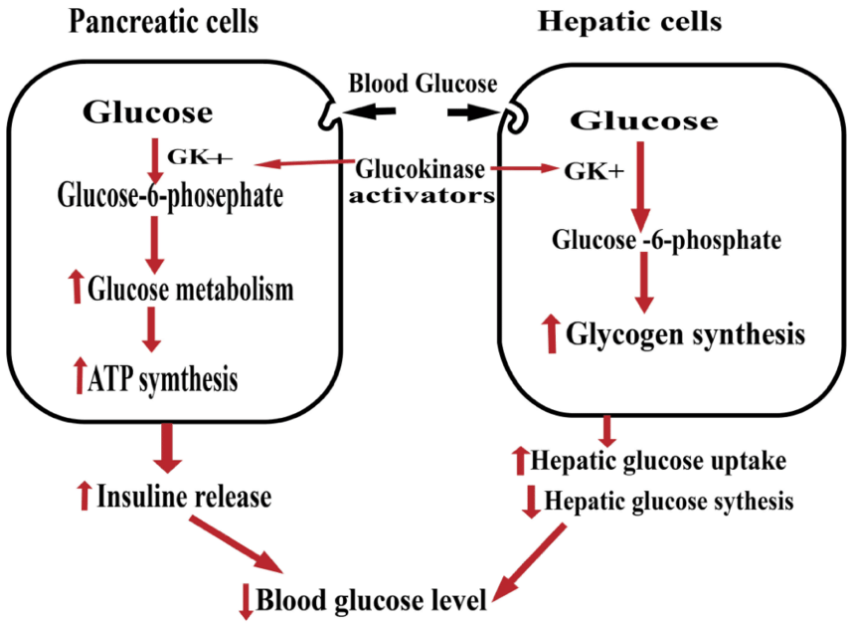
Glucose reabsorption from the glomerular filtrate through a proximal tubule epithelial cell into the blood.

Centrally Acting Drug

- Bromocriptine
- A semisynthetic ergot alkaloid derivative with potent dopaminergic activity
- Activate hypothalamic dopaminergic receptors→dopamine activity ↑ and NE & 5HT activities ↓ → hepatic glucose uptake↓ and insulin resistance ↓
- ADR: N/V, headache, orthostatic hypotension

Dual-acting glucokinase activator

- Dorzagliatin (China FDA approved 2565)
- Monotherapy and as an add-on to metformin (when glycaemic control is inadequate with metformin alone)
- Dual-Acting Glucokinase Activator



- ADR : caused a mild increase in TG and TC, abdominal discomfort, N/V, diarrhea

DRUG	FPG reduction (mg/dL)	HbA1C reduction (%)	Hypoglycemia
Sulfonylureas	60-70	1.5-2	+++
Metformin	60-70	1.5-2	-
Glitazones	35-40	0.5-1.5	-
Meglitinides	13-31	0.6-0.4	+/++
α-glucosidase inhibitors	20-30 (PPG 30-70)	0.7-1	-
Amylin		0.5-1	-
DPP4 inhibitor		0.5-1	+

หลักการเลือกใช้ยารักษาโรคเบาหวาน

1. ผู้ป่วย Type 1 ใช้ อินสุลิน
2. Type 2 เริ่มด้วยการจำกัดอาหาร และออกกำลังกาย เมื่อไม่ได้ผลอาจเริ่มด้วยยาลดน้ำตาลชนิดรับประทานหรืออินสุลิน
3. การใช้ยาลดน้ำตาลชนิดรับประทาน

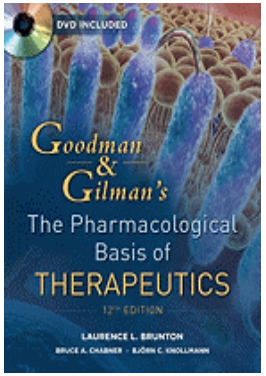
   ผู้ป่วยอ้วน : เริ่มด้วย biguanide ถ้าไม่ได้ผลอาจให้ยาชนิดอื่นร่วมด้วยก่อนเปลี่ยนไปใช้อินสุลิน

   ผู้ป่วยไม่อ้วน : เริ่มด้วย sulfonylurea ชนิดใดชนิดหนึ่ง ถ้าไม่ได้ผล อาจให้ยาชนิดอื่นร่วมด้วย ก่อนเปลี่ยนไปใชอินสุลิน

4. การใช้ยาที่มีกลไกการออกฤทธิ์เหมือนกัน 2 ชนิดร่วมกัน ไม่ทำให้ผลการรักษาดีขึ้น แต่อาจทำให้ความเป็นพิษเพิ่มขึ้น

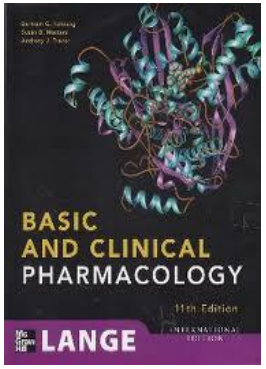
5. ควรประเมินผลการใช้ยาอย่างสม่ำเสมอโดยตรวจระดับน้ำตาลในเลือด (fasting blood glucose, $Hb_{A1\,C}$)

References



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Joel G. Hardman, PhD
Lee E. Limbird, PhD
Alfred Goodman Gilman, MD, PhD

The "Bible of Pharmacology"! The book of choice for hopeful MDs and PharmDs. Very comprehensive and challenging!



Basic & Clinical Pharmacology, 13e, 2015
Bertram G. Katzung, MD (Editor)

Also a great choice for prospective MDs, but overkill for BIMM118 alone (requires strong knowledge of medical terminology)